

Software Requirements Specification for Cooking Management Application - Group 11

Version: 1.0

Date: May 31, 2026

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1. Introduction

1.1 Purpose

This document is for the purpose of describing the software requirements for a Cooking Manager App, which allows users to save recipes either manually or from online sources, plan weekly meals using a calendar, and prepare a grocery shopping list based on the saved recipes/planned meals.

1.2 Scope

This application allows users to simplify meal planning and recipe organization. It allows users to store recipes in a single location, add meals to a calendar for easier planning, and keep a shopping list for needed ingredients. The goal of this app is to reduce the effort needed to prepare meals at home and make cooking seem a less complicated task.

1.3 Definitions/Glossary

- User: A person accessing/interacting with the app to manage recipes, schedule meal plans, and prepare grocery lists.

- Recipe: A collection of information containing a title, list of ingredients, preparation instructions, and optional data like preparation time or serving sizes.
- Recipe Library: A personal collection of recipes imported and saved by the user, which can be searched, filtered, and organized using tags.
- Recipe Import: The process of automatically extracting recipe data from an external source, like a website or article, by parsing the URL and reading structured metadata to fill in the corresponding recipe details.
- Meal Plan: A weekly, calendar based schedule that allows the user to assign meals to specific slots (breakfast/lunch/dinner) to organize and plan future meals.
- Grocery List: A list of ingredients derived from the saved recipes which are needed for meals listed on the weekly meal plan. The list can be either automatically generated or manually created/modified by the user.
- Tag: A user-defined keyword label that can be attached to a recipe for filtering and organization (vegetarian, breakfast, etc.)

1.4 References

None.

2. Overall Description

2.1 Product Perspective

- Frontend - Runs in the user's web browser and displays pages for recipes, meal planning, and grocery lists.
- Backend - Handles login, recipe storage, meal planning logic, and grocery list generation.
- Database - Stores user accounts, recipes, meal plans, and grocery list data.
- Overall - The frontend sends requests to the backend, the backend processes them, and the database stores or retrieves the required information.

2.2 Product Features

- User authentication
- Recipe storage and organization
- Meal planning calendar
- Grocery list generation
- Grocery list editing and management

2.3 User Classes and Characteristics

- Registered users: Can create accounts, save recipes, set meal plans, create and save grocery lists

2.4 Operating Environment

- Web browser for frontend access
- Server running on Windows for backend

2.5 Design and Implementation Constraints

- Should work on latest versions of Chrome, Firefox and Edge browsers
- Passwords/User details must be securely stored using encryption
- Application should accurately save and update user and recipe information

2.6 System Constraints and Dependencies

- Users must have a stable internet connection
 - User data, including credentials, recipes, meal plans, etc. must be securely stored in the system database
 - Application depends on the availability of the backend server/database for operation
 - Recipe import functionality may depend on structure/availability of external websites
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3. Specific Requirements

3.1 Functional Requirements

- **FR1:** User should be able to create an account
- **FR2:** User should be able to log in and log out
- **FR3:** User should be able to add, edit, and delete recipes
- **FR4:** User should be able to organize/categorize recipes
- **FR5:** User should be able to search recipe by name or category
- **FR6:** User should be able to create and modify weekly meal plan in a calendar
- **FR7:** System should generate a grocery list based on planned meals
- **FR8:** User should be able to edit generated grocery list
- **FR9:** User should be able to mark list items as purchased

3.2 Non-Functional Requirements

- **NFR1:** System should support at least 50 users
- **NFR2:** User passwords must be safely encrypted before storage
- **NFR3:** System should provide a consistent UI across all pages and show meaningful error messages for invalid inputs

3.3 Interface Requirements

- **Authentication Interface:** Login, registration, and account management functions
- **Web Interface:** Web browser interface providing access to Recipe Library, Meal Planner, Grocery List, and Account Settings
- **Meal Planning Interface:** Calendar-based meal scheduling view
- **Grocery List Interface:** Interactive shopping list view

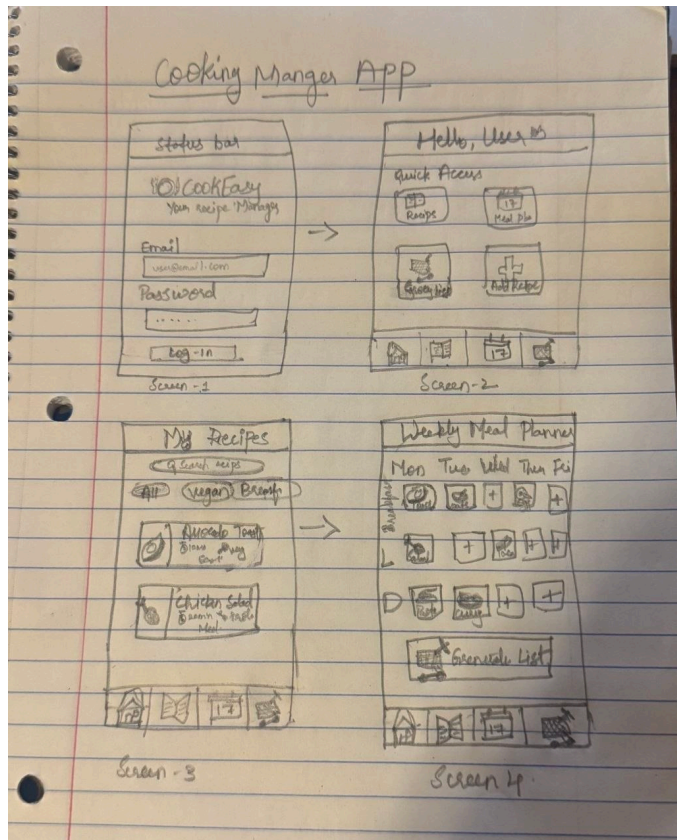
3.4 Data Requirements

- **User Information:** Username, Password, Email
- **Recipe Information:** Recipe Number/ID, Name, Category, Ingredients, Instructions, Preparation Time

- Meal Plan Information: Plan ID, Date, Assigned Recipe, Time Slot (Breakfast/Lunch/Dinner), Recipe ID
- Grocery List Information: Item ID, Item Name, Quantity Needed, Purchase Status

3.5 User Interface Sketches

- **Concept A - Dashboard-Based Layout**



- **Concept B - List-Based Navigation**

UI Sketch 1 - login page

Cooking manager

Email:
Password:

login button

create account

UI Sketch 2 - Recipe library

Header: Recipes | meal planner | Grocery list

Search bar: search recipes

Recipe card 1

- image place holder
- recipe name
- category

Recipe card 2

- image Placeholder
- recipe name
- category

meal planner

mon	tue	wed	thurs	fri	sat	sunday
B: [Recipe]	[Recipe]	[Empty]	...			
L: [empty]	[Recipe]	[Recipe]	...			
D: [Recipe]	[Empty]	[Recipe]	...			

UI sketch 3

meal planner

weekly calendar

UI Sketch 4 - Grocery list

Grocery list

Chicken Breast
Olive oil
Tomatoes (4)
Pasta (4 box)

Add Item

Clear Purchased